Course Syllabus

Sky, Time and Telescopes — an open astronomy course for learners aged 16+

Open to	everyone aged **16+**: students, college learners, teachers, adults — no prior astronomy needed
Maths needed	Class 10 level: ratios, simple equations, a calculator for sin/cos/tan. Logarithms are introduced when needed, and every calculation has a worked example first.
Format	27 classroom sessions (~90 min) + 5 observing nights + 1 telescope workshop
Duration	October 2026 – early April 2027 (India's best observing season)
Units	1. Motion of the Sky and Time-Keeping · 2. Sky, Stars and Constellations · 3. Magnitudes, Telescopes and Catalogues

Why this course?

Look up on a clear night and you are looking at the oldest science there is. The same sky set the dates of Makar Sankranti and Eid, named our weekdays, guided sailors and inspired Aryabhata, Ptolemy and al-Sufi. This course teaches you to **read the sky like a clock and a map**, to understand the calendars and festivals around you, and to **use binoculars and telescopes with confidence**. It is designed to be **interesting and full of knowledge, but never so hard that anyone gives up**. Every unit mixes stories and history (especially Indian and other cultural traditions), memory and pattern work, simple practical calculations with worked examples, and real observing.

Difficulty is marked everywhere: ★ remember ★★ apply ★★★ challenge (optional bonus). Anyone who completes the ★ and ★★ work will pass well; the ★★★ questions stretch olympiad-minded students.

Learning outcomes

By the end of the course you will be able to:

1. **explain** the daily and yearly motions of the sky, the seasons, the Moon's phases and eclipses;
2. **describe and compare** the Hindu (Panchang), Islamic (Hijri) and Gregorian calendars, and carry out simple date and time conversions;
3. **use** the celestial sphere and the Alt–Az and RA–Dec coordinate systems to predict where and when objects are seen from India;
4. **identify by sight** at least 15 constellations/asterisms and 20 bright stars, with their Indian names, and **star-hop** in every season;
5. **calculate** brightness ratios, absolute magnitudes and distances with the magnitude system;
6. **choose, set up and use** a telescope, and calculate magnification, field of view, exit pupil, resolution and limiting magnitude;
7. **use** star and deep-sky catalogues to plan an observing session.

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Course map

Unit	Lectures	Assignments
1. Motion of the Sky and Time-Keeping	1.1 Earth's rotation & diurnal motion · 1.2 Seasons, equinoxes, solstices, Makar Sankranti, Zero Shadow Day · 1.3 The Moon, phases & eclipses · 1.4 Hindu time-keeping: weekdays, tithis, nakshatras, Panchang · 1.5 The Islamic Hijri calendar · 1.6 Gregorian calendar, leap years & IST	1A Sky motion, seasons, Moon & eclipses · 1B Calendars & time-keeping · 1C Project: Moon diary & shadow stick
2. Sky, Stars and Constellations	2.1 The celestial sphere · 2.2 Coordinate systems · 2.3 Constellations: history & the modern 88 · 2.4 Asterisms, zodiac & bright stars · 2.5 Stargazing field guide for India	2A Sphere & coordinates · 2B Picture quiz · 2C Project: constellation hunt
3. Magnitudes, Telescopes and Catalogues	3.1 Apparent & absolute magnitudes · 3.2 Telescope types · 3.3 Telescope calculations · 3.4 Setting up a telescope (workshop) · 3.5 Star & deep-sky catalogues	3A Magnitudes · 3B Telescope calculations · 3C Capstone: observing plan
Final	—	Final Assessment (all units, including picture identification)

Lecture notes are handed out **after each lecture**; each note builds only on what has already been taught.

Session-by-session schedule

Dates are suggestions. Observing nights fall near New Moon (dark skies) or First Quarter (Moon viewing); each has a cloudy-night backup (indoor Stellarium session).

Wk	Dates	Session	Sky this week / event
1	5–11 Oct 2026	Orientation · L1.1 Earth's rotation	New Moon 10 Oct
2	12–18 Oct	L1.2 (part 1) Seasons, ecliptic, equinoxes	**Observing Night 1 (Sat 17 Oct):** Moon near First Quarter + Saturn; start the Moon Diary (1C)
3	19–25 Oct	L1.2 (part 2) Makar Sankranti, precession, Zero Shadow Day	Orionids ~21 Oct
4	26 Oct–1 Nov	L1.3 The Moon, phases & eclipses · *1A out*	Full Moon 26 Oct
5	2–8 Nov	L1.4 (part 1) Weekdays, tithis, festivals	Diwali on the Amavasya around 8 Nov — discuss why!
6	9–15 Nov	L1.4 (part 2) Months, Adhik Maas, nakshatras, time units	**Observing Night 2 (Sat 14 Nov):** autumn sky, Andromeda Galaxy, Saturn
7	16–22 Nov	L1.5 The Islamic Hijri calendar · *1A due*	Leonids ~17 Nov
8	23–29 Nov	L1.6 Gregorian calendar, leap years, IST · *1B out*	
9	30 Nov–6 Dec	Unit 1 review + quiz game · *1B, 1C due*	
10	7–13 Dec	L2.1 The celestial sphere	**Observing Night 3 (Sat 12 Dec):** winter constellations + Geminids (peak 13–14 Dec)
11	14–20 Dec	L2.2 (part 1) Alt–Az & RA–Dec	
12	21–27 Dec	L2.2 (part 2) Meridian altitude, LST · *2A out*	Winter solstice 21–22 Dec
13	28 Dec–3 Jan 2027	*Winter break* (self-study 2A; start 2C observing)	
14	4–10 Jan	L2.3 Constellations: history & the 88	**Observing Night 4 (Sat 9 Jan):** Orion, Winter Hexagon, Orion Nebula
15	11–17 Jan	L2.4 (part 1) Asterisms & bright stars · *2A due*	Makar Sankranti 14–15 Jan — link back to L1.2!

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Wk	Dates	Session	Sky this week / event
16	18–24 Jan	L2.4 (part 2) Zodiac, Indian star names, star vs planet	
17	25–31 Jan	L2.5 Stargazing field guide · **2B picture quiz (in class)**	
18	1–7 Feb	Unit 2 review · *2C due* · L3.1 (part 1) Apparent magnitudes	New Moon 6 Feb
19	8–14 Feb	L3.2 Telescope types	**Telescope Workshop (Sat 13 Feb), L3.4:** daytime setup, then the Moon + Jupiter & Mars at opposition in Leo
20	15–21 Feb	L3.1 (part 2) Absolute magnitudes & distances · *3A out*	
21	22–28 Feb	L3.3 (part 1) Magnification, exit pupil, field of view	
22	1–7 Mar	L3.3 (part 2) Resolution, limiting magnitude, imaging · *3A due, 3B out*	**Observing Night 5 (Sat 6 Mar):** deep-sky night (New Moon 8 Mar)
23	8–14 Mar	L3.5 Star & deep-sky catalogues · *3C out*	
24	15–21 Mar	Unit 3 review · *3B due*	March equinox ~20 Mar
25	22–28 Mar	**Final Assessment** (2 hours)	
26	29 Mar–4 Apr	Capstone presentations (3C) + certificates	Optional Messier Marathon, night of Sat 3 Apr (New Moon 7 Apr)

Assessment

Component	Items	Weight
Written assignments	1A, 1B, 2A, 2B, 3A, 3B	**40%**
Observation projects	1C, 2C, 3C (capstone)	**30%**
Final assessment	2-hour paper, all units, incl. picture identification	**20%**
Participation	observing nights, workshop, class activities	**10%**

Certificate of Completion: overall ≥ 50% and at least 3 of the 5 observing nights (or approved make-ups). **Distinction:** overall ≥ 80%. ★★★ bonus marks are added on top within the same assignment.

Every written assignment follows the same pattern: fill-in-the-blanks / MCQ (★), matching and patterns (★), short answers (★★), calculations with a worked example and formula box (★★), a night-sky task (★★) and an optional bonus (★★★).

What you need

A notebook and pencil; a basic scientific calculator; a **red torch** (red cellophane over a torch is fine); the free planetarium program **Stellarium** (`https://stellarium.org`) on a phone or computer; optionally 7 × 50 or 10 × 50 binoculars. Telescopes are provided for the workshop and observing nights.

⚠ Rules for observing nights

1. **Never look at the Sun** through any optical instrument without a certified solar filter.
2. Red light only after dark — no white phone screens near the telescopes.
3. Ask before touching a telescope; do not lean on tripods.
4. Laser pointers only by instructors, briefly, never towards aircraft or people.
5. Stay with the group; minors need a guardian's consent for night sessions.

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Free and public reading

- OpenStax, *Astronomy 2e* (free textbook, CC BY 4.0) — https://openstax.org/details/books/astronomy-2e
- NCERT *Physics* Class XII, Part II, Ch. 9 "Ray Optics and Optical Instruments" — https://ncert.nic.in/textbook.php
- IAU constellation charts — https://www.iau.org/public/themes/constellations/; Stellarium User Guide — https://stellarium.org
- Widely available books: H. A. Rey, *The Stars: A New Way to See Them*; G. Consolmagno & D. M. Davis, *Turn Left at Orion*; T. Dickinson, *NightWatch*; S. Balachandra Rao, *Indian Astronomy: An Introduction*.
- Olympiad syllabi that informed the course: Junior IOAA (https://ioaastrophysics.org/junior-ioaa/junior-ioaa-syllabus); INAO/NSEA (HBCSE–IAPT, https://olympiads.hbcse.tifr.res.in).